



Research paper

SiamTU: Hyperspectral video tracking against appearance changes using improved SiameGAT with adaptive template updating

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ABSTRACT

Siamese-based tracking algorithms are gaining popularity in hyperspectral tracking because of their effectiveness in feature matching. Nevertheless, these techniques require refinement to handle significant variations in the target's appearance, including deformation, occlusion, and small size, more effectively. Therefore, we propose the SiamTU (SiameGAT with Template Updating) tracker, which includes band selection, an improved graph attention based Siamese model, and an adaptive template update mechanism, using public hyperspectral videos. Initially, a band selection technique employing the neighborhood group normalized filter is utilized to obtain three high-significance bands. The resultant synthesized image is then used as the input for the tracker. Following this, a module for efficient feature refinement is developed to improve the features that have been encoded. The model has the ability to simultaneously concentrate on both local details and overall structures by extracting features from the input at various scales. Moreover, an embedded template update network enhances the SiamTU's capability referring to appearance changes effectively. Results from tests on the hyperspectral dataset reveal that SiamTU is more effective than comparable algorithms, attaining a success rate of 0.617 and a precision value of 0.939. The code will be accessible at <https://github.com/ctb2/SiamTU>.

1. Introduction

Object tracking seeks to estimate the state of the target in real-time, but tracking performance declines substantially in difficult situations, such as occlusion and scale variation. Recently, hyperspectral videos containing rich spectral information have been used to address these issues. In particular, the Hyper-Spectral Tracking (HST) [1–5] algorithms perform well when faced with false objects of the same color. However, these approaches also face the difficulties of data redundancy and low real-time performance, which needs to be solved using Hyper-Spectral Videos (HSVs).

Recently, HST techniques have been divided into two main categories: those based on Correlation Filters (CF) and those relying on Deep Learning (DL). The CF leverages both spectral and spatial attributes of the target, by training correlation filters to forecast the target's location in future frames [6–9]. Chen et al. [10] introduce an efficient feature extraction technique utilizing spatial-spectral convolution kernels to enable real-time processing. Xiong et al. [11] introduce Mterial-based Hyperspectral Tracking (MHT), which capture material appearances in hyperspectral images. However, this method requires enhancements to perform well under different lighting conditions. Hou

et al. [12] introduce a tensor-sparse correlation filter enhanced with spatial spectral regularization to enhance feature differentiation and scale adaptability. This approach merges tensor processing and a two-dimensional adaptive scale search method. The manually designed feature extraction models mentioned above struggle to adjust effectively in changing environments.

Subsequently, DL models that have demonstrated exceptional visual tracking performance are now being used in the HST domain. Uzkent et al. [13] integrate CF with deep features for the purpose of tracking from an aerial perspective in HSVs, yet this approach overlooks spectral information. To tackle this problem, a hyperspectral target-aware module is incorporated into the Siamese module, integrating features extracted from both visual and hyperspectral images [14]. The Band Attention-aware Ensemble Network (BAE-Net) [15] processes three-channel images with a deep color network and uses ensemble learning to combine weak trackers for localization. However, redundant spectral information potentially hinders the tracking precision. To improve the efficiency of the algorithm, BS-SiamRPN (SiamRPN based on Band Selection) [16], GOMT (Genetic Optimization and Multi-features Tracking) [17], ADSiamRPN (Anti-Deformation SiamRPN) [18] and

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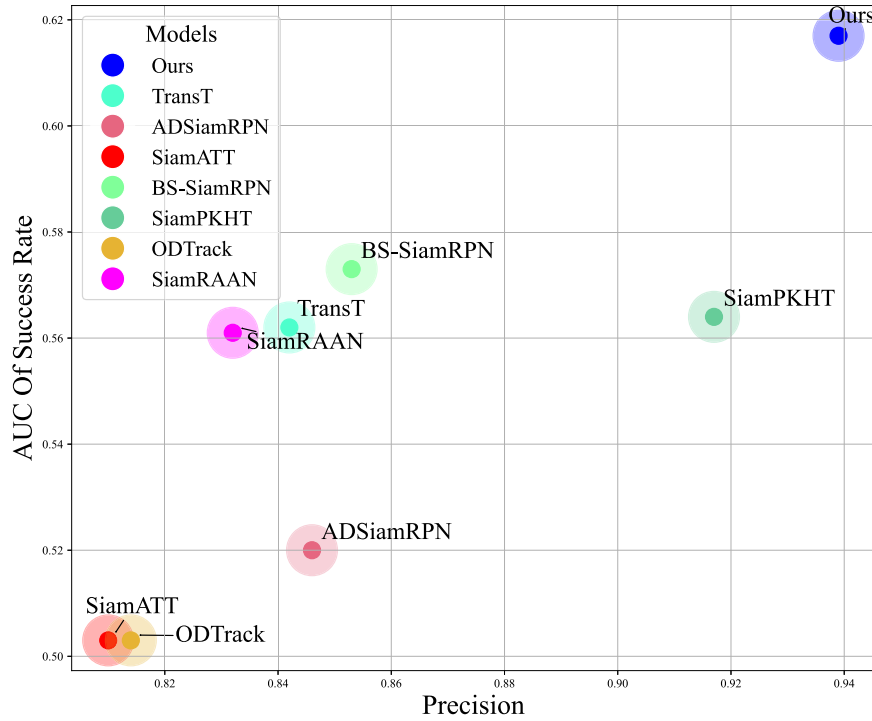


Fig. 1. The scatter plot of the tracking performance. As points approach the upper right corner, the model's performance improves.

SiamPKHT (Siamese with Pyramid attention and Knowledge distillation) [19] employ genetic optimization techniques to choose specific bands. In addition, Zhao et al. [20] propose a spectral matching dimensionality reduction approach, which differentiates pixels within the search area by applying two predetermined thresholds. However, the precision is greatly affected by the target appearance change. On the one hand, the target template does not adapt well to changes in its appearance, a challenge similarly observed in RGB tracking where fixed templates struggle to cope with dynamic visual changes. On the other hand, the performance of feature extraction module needs to be further enhanced.

As a result, we propose a Siamese-based algorithm named SiamTU (SiamGAT with Template Updating) to deal with appearance changes in HSVs. The SiamTU consists of SiamGAT (Siamese Graph Attention Tracking), Efficient Multiscale Attention (EMA) and a template UpdateNet (Update Network). First, band selection based on the Neighborhood Group Normalized Matched Filter (NGNMF) is used to reduce the image dimension, retaining valuable spectral information. Then, the EMA organizes the reconstructed channels into several sub-features, ensuring that within each group, the distribution of spatial semantic features is consistent. In addition, UpdateNet is embedded to enhance the template features, incorporating update constraints to discard some poorly tracked frames. Experimental results show that the proposed SiamTU tracker is highly precise in public infrared hyperspectral videos referring to all attributes. The scatter plot of the tracking performance is shown in Fig. 1. Key contributions include the following.

- The proposed SiamTU obtains high precision in the dataset named IMEC25 (Israel Materials Engineering Conference with 25 bands), demonstrating its effectiveness in tracking a hyperspectral target with appearance changes.
- The lightweight neighborhood group-matched filter that retains valuable spectral information helps to obtain an optimal combination of bands and increase the corresponding real-time performance.
- The collaborative EMA model that integrates multiscale attention with spatial information contributes to building a strong representation, thus improving the hit rate of feature matching that refers to the Siamese model.

- In addition, the designed template update mechanism including the adaptive UpdateNet and the selection model makes the template adapt well to the target appearance changes and cluttered background.

This paper's core content is outlined as follows: Section 1 provides an introduction. Section 2 discusses related work concerning the proposed method. Section 3 details the proposed tracking algorithm. Section 4 examines the experimental results, and Section 5 draws conclusions.

2. Related work

To improve object tracking performance referring to hyperspectral information, we design a Siamese-based tracker that offers the advantages of band selection and template update. This section offers a concise overview of hyperspectral video tracking and Siamese trackers, emphasizing the rationale behind the proposed algorithm.

2.1. Hyperspectral video tracking

Hyperspectral images contain a wealth of spectral bands, which are instrumental in differentiating objects with comparable visible spectrum reflectance. This significantly increases the discriminability of these images and opens up further possibilities for target tracking applications.

Early hyperspectral video tracking relied primarily on hand-made features to represent objects. To our knowledge, Banerjee et al. [21] first use hyperspectral information to address the challenging problems encountered in video surveillance. Qian et al. [8] propose a CF-based hyperspectral tracker using the three-dimensional convolutional feature, obtaining spectral information around the target. Here, redundant information requires high performance of the classifier. Both the material feature model [11] and the adaptive spatial spectral model [22] address this issue well [11]. However, manually crafted features can lead to drift in tracking when modeling the appearance of the target [23–26].

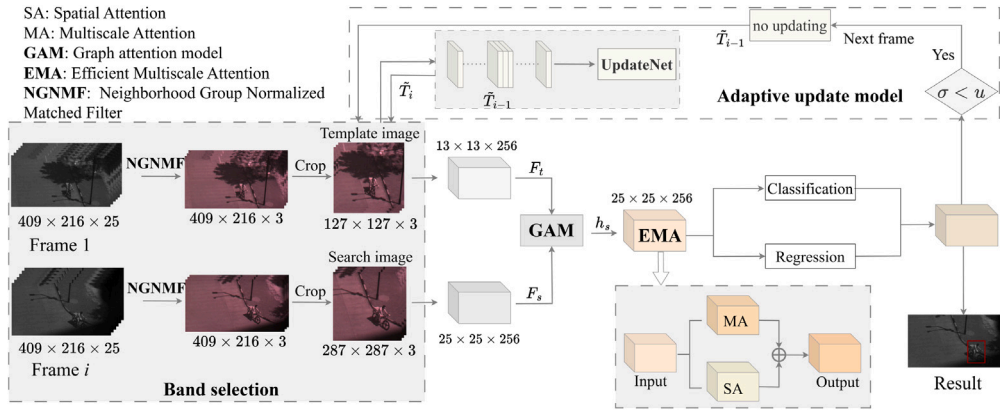


Fig. 2. The overview of SiamTU.

Subsequently, hyperspectral video trackers based on DL methods have gradually emerged. Xiao et al. [27] introduce a two-branch Siamese network using all spectral bands, which may introduce interference information. Moreover, Siamese with Band Attention Grouping (SiamBAG) [28] assigns weights to the bands through band attention and transforms hyperspectral images into several pseudo-color images for training. They also merge an end-to-end network with the Siamese network [29] and incorporate a prediction module that evaluates the significance of the band. Recently, transformers have gradually gained popularity [30,31], including TransT (Transformer Tracking) [32] and ODTrack (Online Dense Temporal Token Learning for Visual Tracking) [33]. Based on it, Wang et al. [34] propose the Spectral-Spatial Transformer Fusion Feature (SSTFT), which can capture the interactions of self-context and cross-context features. The SPTrack [35] converts hyperspectral information into a single-channel heat map and acquires combined features from both the heat map and the three-channel images. However, the computational difficulty increases, and the precision is greatly affected by the target appearance change. Therefore, we take advantage of band selection and focus on enhancing the efficiency of the Siamese method.

2.2. Siamese trackers

Siamese trackers store the target appearance in an explicit template, eliminating the need for online updates. Both branches of the Siamese network are processed through the identical feature extractor, independently acquiring the template features and the features of the search area. Two feature maps are subsequently compared through cross-correlation, allowing the target's position to be identified from their similarity.

Pioneering research on SiamFC (Fully Convolutional Siamese) [36] lays the foundation for subsequent Siamese tracking. Inspired by the region proposal network, SiamRPN (Siamese Region Proposal Network) [37] shares parameters between the classification and regression tasks. DASiamRPN (Distractor-aware SiamRPN) [38] builds on the former by adding a large number of positive samples and enriching semantic background information to improve tracking performance. However, the feature extraction used in these algorithms is constrained by spatial invariance. SiamRPN++ [39] breaks this limitation by employing a spatially aware sampling strategy. SiamATT (Siamese ATtention) [40] adds an attention branch based on SiamRPN++. SiamRAAN (Siamese Residual Attentional Aggregation Network) [41] performs implicit template feature updates during the template and search feature extraction process, essentially enhancing features. To overcome the drawback of anchor, several anchor-free trackers, including methods such as SiamFC++ [42], SiamCAR (Siamese Classification And Regression) [43], and SiamBAN (Siamese Box Adaptive Network) [44], have been introduced. These approaches reduce the trackers' reliance

on hyperparameters to a certain degree. SiamAttn (Siamese Attention networks) [45] presents an innovative Siamese attention approach designed to improve the tracking ability to learn features. SiamGAT [46] employs attention mechanisms to link target features with template features, thereby enhancing the flow of information related to the target. SiamCAT [47] employs a dual Siamese network model, which resolves the issues of varying channel numbers in hyperspectral data and the insufficient utilization of spectral features through a channel adaptive module and a guided learning attention module. SiamOHOT [48] also adopts a dual Siamese network structure for real-time airborne hyperspectral object tracking. In addition to that, the aforementioned method struggles to effectively handle challenges such as deformation and occlusion. This serves as a rationale for the proposed approach, which employs an efficient template update mechanism.

3. The SiamTU tracker

This section gives the details of SiamTU (Fig. 2), which consists of band selection, the improved SiamGAT model and the template update model.

3.1. Band selection

Deep learning techniques has been widely applied in the preprocessing of hyperspectral images, but it is associated with issues such as time consumption, extensive hyperparameters, and high computational costs. This section employs a lightweight band selection technique rooted in the normalized matched filter for neighborhood grouping [49], which incorporates both band grouping and ranking, as presented in Fig. 3, allowing for a more straightforward extraction of the spectral-spatial features of hyperspectral images. The NGNMF, whose main objective is to obtain K bands with high weights in the K groups, reduces spectral redundancy while preserving relevant spectral information.

To be specific, the final band set $Y = \{Y_1, \dots, Y_k, \dots, Y_K\}$ can be obtained according to the corresponding band group $X = \{x_1, \dots, x_k, \dots, x_K\}$, where x_k refers to the original k th group with a fine partition algorithm [50]. The spectral information in each band group x_k is highly correlated, while the correlation between different groups is relatively low. Moreover, the group x_k can be represented by

$$x_k = \begin{cases} (X^{(k-1) \times [B/K] + 1}, \dots, X^{k \times [B/K]}) & k \times [B/K] \leq B \\ (X^{(k-1) \times [B/K] + 1}, \dots, X^B) & \text{else} \end{cases} \quad (1)$$

where X^B represents the B th band within X , and $[B/K]$ refers to the smallest integer greater than or equal to B/K . Here, $B = 25$ and $K = 3$, x_1 , x_2 and x_3 can be represented by $X^1 \sim X^9$, $X^{10} \sim X^{18}$, and $X^{19} \sim X^{25}$, respectively.

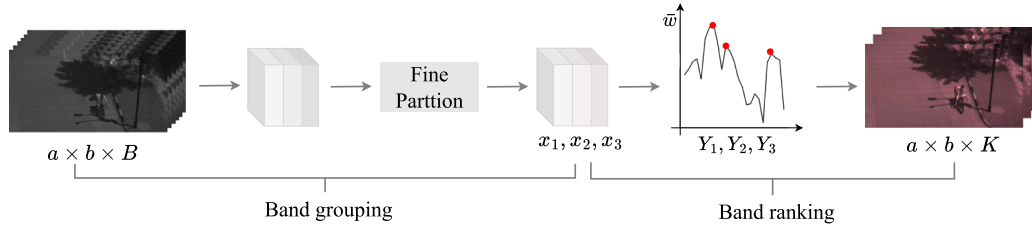


Fig. 3. The flowchart of NGNMF-based band selection.

Subsequently, Y_k is determined by the weights of each band in the group x_k using the mean of the absolute weight vector.

$$\bar{w}_k = \frac{1}{a \times b} \sum_{l=1}^{a \times b} |w_k^l| \quad (2)$$

where $\bar{w}_k$ denotes the weight vector linked to the band group x_k possessing a width a and a height b . The w_k^l , which indicates the weight referring to the l th pixel vector, can be represented as

$$w_k^l = \kappa C^{-1} (x_k^l - m_k) \quad (3)$$

where m_k , C and κ denote the mean of all x_k^l , the covariance matrix and the normalization constant.

3.2. The siamese baseline

With the selected bands, we improve SiamGAT using efficient multiscale attention and adaptive UpdateNet, effectively tracking the hyperspectral target, as shown in Fig. 2.

3.2.1. SiamGAT

In SiamGAT [46], reliable information from the target template can be transferred to the search area with a part-to-part correspondence. Specifically, SiamGAT consists mainly of three components: a feature extractor, a Graph-Attention Module (GAM) to embed target information, and a network for classification and regression.

The bands $Y = \{Y_1, Y_2, Y_3\} \in R^{a \times b \times 3}$ chosen are employed to derive both the template and search features, with their respective node sets denoted as V_t and V_s . Furthermore, the complete bipartite graph G is given by

$$G = (V_s \cup V_t, \{(u, v) \mid \forall u \in V_s, \forall v \in V_t\}) \quad (4)$$

Let h_s^u and h_t^v represent the nodes u and v , respectively. The final feature h_s^u referring to each node u can be expressed as

$$\hat{h}_s^u = \mathbf{R}(V_u \parallel (W \times h_s^u)) \quad (5)$$

where W , $\mathbf{R}$ and $\parallel$ denote the linear transformation matrix, the rectified linear function and the concatenation operation. Parallely compute all $\hat{h}_s^u$ ($\forall u \in V_s$), obtaining the final response map h_s . The aggregated feature V_u of node u can be represented by

$$V_u = \sum_{v \in V_t} (\mathbf{S}_1(e_{uv}) W \times h_t^v) \quad (6)$$

where the inner product e_{uv} acts as the similarity measure between the search features and the template features. Here, $\mathbf{S}_1$ represents the softmax function.

SiamGAT employs the classification-regression head network from SiamCAR [43] to function as its tracking head. In Fig. 2, the classification branch is responsible for predicting the category at each position, whereas the regression branch computes the size of the bounding box for the target located at that position. A central branch is incorporated into the classification branch to eliminate anomalous prediction boxes. Therefore, the total loss $\mathcal{L}$ is expressed as

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{cen} + \lambda_2 \mathcal{L}_{reg} \quad (7)$$

where $\mathcal{L}_{cls}$, $\mathcal{L}_{cen}$ and $\mathcal{L}_{reg}$ respectively represent the classification loss, the center-ness loss, and the regression loss, with λ_1 and λ_2 being constants. More details can be found in [43]. During model training, we set $\lambda_1 = 1$ and $\lambda_2 = 3$.

3.2.2. Efficient multiscale attention

The existing attention mechanisms mostly require dimensional reduction to model cross-channel relationships, which may affect the deep expression of features. Therefore, the Multiscale Attention (MA) [51] model is used without the need for dimensional reduction, as shown in Fig. 4. Initially, it reconfigures part of the input channel by converting it into the batch dimension, thereby structuring them into sub-feature groups within the channel dimension. Then, the channel weights are adjusted within each branch. Ultimately, the features are produced via cross-dimensional interactive aggregation.

However, the above channel attention may cause side effects in deep visual representation extraction due to dimensionality reduction. Thus, Spatial Attention (SA) is embedded to help the model improve feature representation for key regions. The SA output is fused with the MA output to obtain the final result O .

$$O = \mathbf{S}(\mathbf{C}(\mathbf{MA}(h_s) + \mathbf{SA}(h_s))) \quad (8)$$

where $\mathbf{MA}$, $\mathbf{SA}$, $\mathbf{C}$, $\mathbf{S}$ denote MA, SA, convolution and sigmoid functions, respectively.

Within the MA module, the function $\mathbf{G}$ partitions the input feature map into g groups along the channel dimension. In addition, three parallel branches are used to extract the channel weight of the grouped feature maps. Then, $\mathbf{MA}(h_s)$ can be expressed as

$$\mathbf{MA}(h_s) = \mathbf{G}(h_s) \times Q \quad (9)$$

where Q denotes the channel weight that can be expressed as

$$Q = \mathbf{D}(\mathbf{P}(d_{12}), d_3) + \mathbf{D}(\mathbf{P}(d_3), d_{12}) \quad (10)$$

where $\mathbf{D}$ and $\mathbf{P}$ denote operations of the matrix dot product and the 2-D global average pooling, respectively. The outputs of the initial two branches are represented by d_{12} , while d_3 signifies the output of the third branch.

Subsequently, SA can be defined as

$$\mathbf{SA}(h_s) = h_s \times \mathbf{S}(\mathbf{C}(\mathbf{M}(h_s) + \mathbf{A}(h_s))) \quad (11)$$

where $\mathbf{M}$ represents maximum pooling and $\mathbf{A}$ signifies average pooling, respectively.

Fig. 5 illustrates the experimental results, which demonstrate the effectiveness of the attention mechanism. In the first column, the target size is small and the information is weak. With EMA, the target information is improved and the corresponding areas in the response map (third column) are highlighted.

3.3. Adaptive template update mechanism

3.3.1. The adaptive UpdateNet

The Siamese model utilizes the initial frame as a static template image for tracking. However, it does not adapt well to the target's

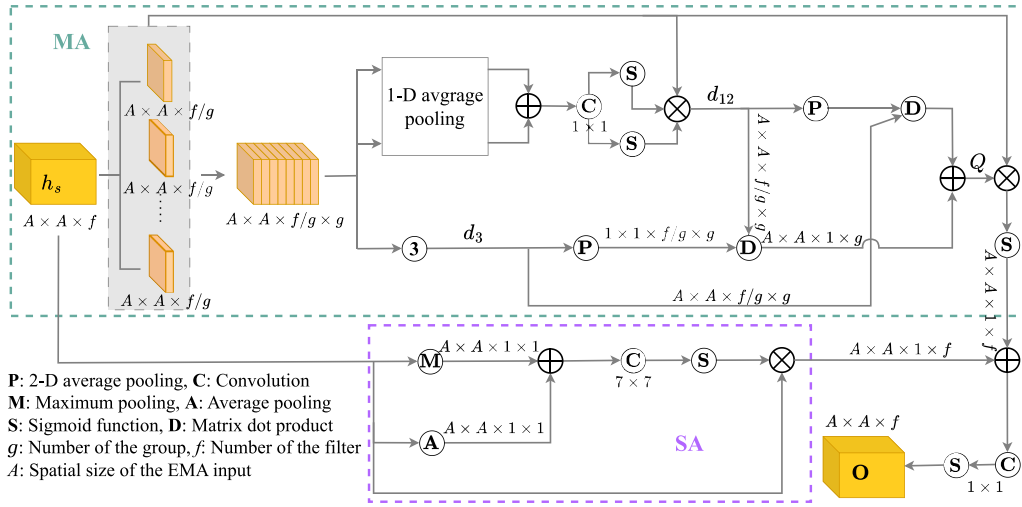


Fig. 4. The EMA module. It comprises two sections, namely MA and SA.

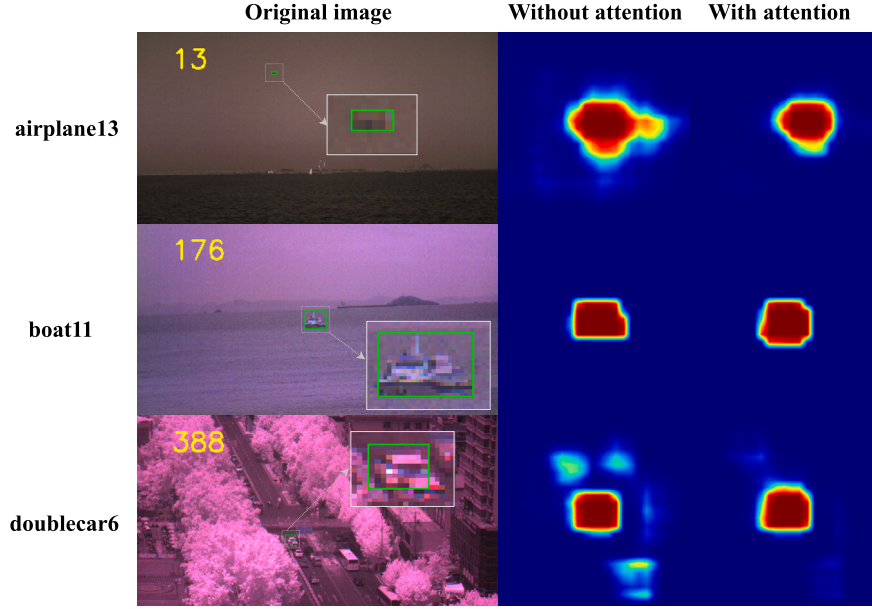


Fig. 5. The effectiveness of the EMA module. The tracker utilizing EMA shows specific expertise in improving the details of small targets.

deformation in later frames. Consequently, the proposed tracker incorporates an adaptive template update mechanism, as illustrated in Fig. 6. The corresponding result T_i^* is defined as

$$T_i^* = \varphi(T_0^{GT}, \tilde{T}_{i-1}, T_i) \quad (12)$$

where φ denotes the feature extractor. $\tilde{T}_{i-1}$, T_0^{GT} , and T_i are the accumulated template, the initial template frame, and the present frame, respectively. The result T_i^* along with T_0^{GT} is entered into the UpdateNet network, which outputs the updated accumulated template $\tilde{T}_i$.

$$\tilde{T}_i = \phi(T_i^*, T_0^{GT}) \quad (13)$$

where ϕ represents the UpdateNet module.

UpdateNet highlights the significance of the initial frame template and incorporates feature information from both the accumulated template and the current frame template. This integration improves the effectiveness of object tracking to some extent, as illustrated in Fig. 7.

3.3.2. The template selection method

For the above UpdateNet module, issues such as target deformation and occlusion can lead to the accumulation of incorrect information in the template. In other words, feeding the template referring to each frame template into the update network without essential filtering might reduce tracking performance in certain scenarios. To address this, a template selection model is designed with a preset threshold u .

$$\tilde{T}_i = \begin{cases} \phi(T_i^*, T_0^{GT}) & \sigma > u \\ \tilde{T}_{i-1} & \sigma \leq u \end{cases} \quad (14)$$

where the classification score σ , which relates to the peak value m_{max} of the associated response map, can be denoted by

$$\sigma = \frac{m_{max}}{\gamma m_{max} + (1 - \gamma) \bar{m}_{i-1}} \quad (15)$$

where $\bar{m}_{i-1}$ represents the mean of the related response map in the frame $(i - 1)$, with γ being a predetermined constant. Generally, it indicates that the new template delivers incorrect information if the score σ drops under the threshold u . As a result, the frame's template is eliminated.

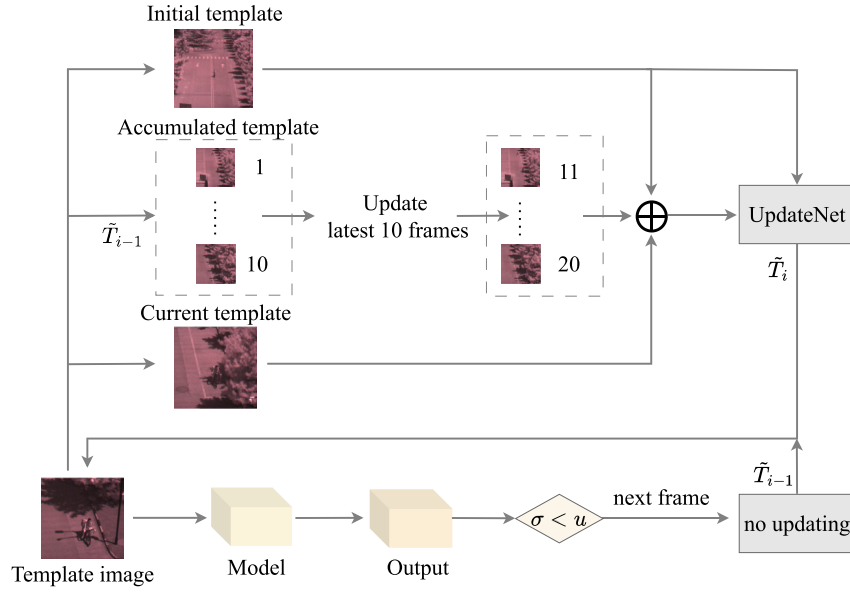


Fig. 6. The flowchart of the adaptive TU mechanism. The numbers indicate the frame numbers of the images.

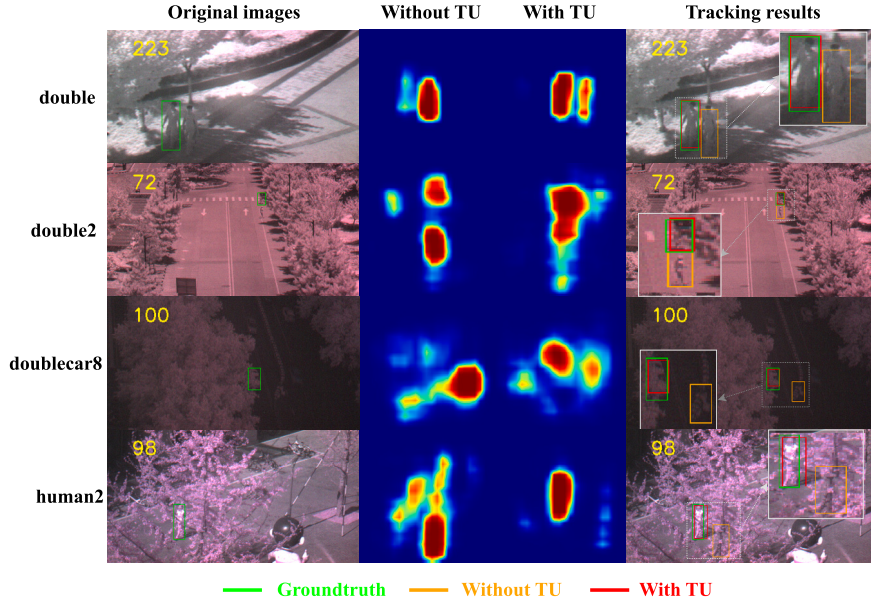


Fig. 7. The effectiveness of the TU module. The tracker employing TU demonstrates particular skills in handling similar objects and occlusion interference.

Subsequently, the appearance of the target can vary greatly in a few seconds, indicating that it is unnecessary to retain accumulated templates far from the current frame. Thus, we combine UpdateNet with a template accumulation mechanism referring to the update interval n . Specifically, when the number of stored templates reaches n , the oldest one is discarded in the subsequent frame, and the accumulation process begins again with the current template. Regularly refreshing the collection of templates in this way ensures that the matching templates maintain elevated confidence levels, thus improving tracking efficiency.

4. Experiments and analysis

Section 4.1 provides a detailed explanation of the experimental setup, encompassing both the dataset and the parameter. Sections 4.2 and 4.3 provide analyses of existing object tracking methods, focusing on qualitative and quantitative aspects, respectively. The advantages and limitations of the proposed algorithm are also discussed. Furthermore, in Section 4.4, we conduct ablation studies to illustrate

the contribution of each component to object tracking in infrared hyperspectral videos.

4.1. Experiment setup

In this paper, the IMEC25 dataset¹ [52] is used for training and testing. Specifically, it comprises 135 annotated video sequences, with 55 sequences (14 000 frames) used for model training and 80 sequences (14 060 frames) for algorithm evaluation. Each image has a resolution of 1080×2045 pixels, with the information from 25 spectral bands stored as single-channel grayscale images. All data were captured in visible and near-infrared (680–960 nm). Fig. 9 shows the band diagram of the dataset. In addition, this dataset is characterized by diverse targets, rich scenes, and high complexity, making it a valuable resource

¹ Available at: <https://github.com/Chenlulu1993/HOMG>.

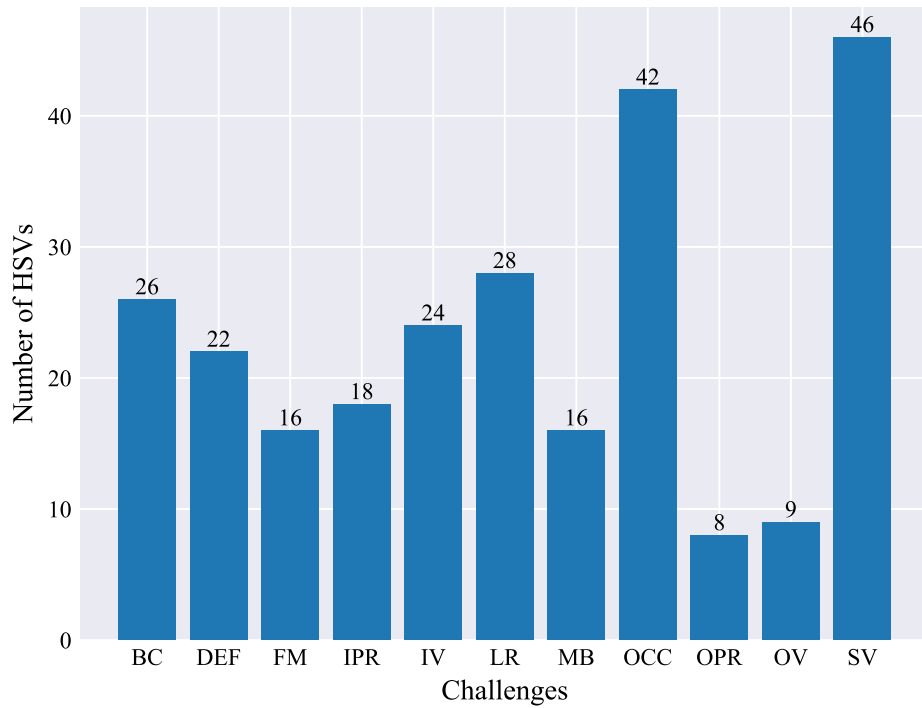


Fig. 8. The bar chart of the HSV number that refers to 11 challenges.

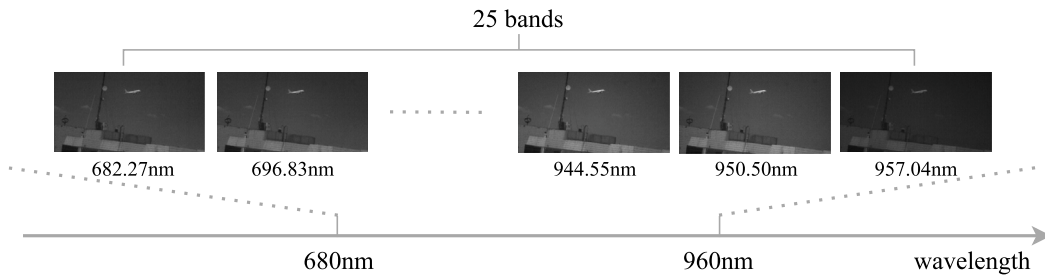


Fig. 9. Band diagram of a frame image in IMEC25 dataset. This dataset falls within the visible and near-infrared spectrum.

for tracking research. The sequence presents 11 challenging attributes such as Illumination Variation (IV), Out-of-Plane Rotation (OPR), Scale Variation (SV), Low Resolution (LR), In-Plane Rotation (IPR), Occlusion (OCC), Deformation (DEF), Fast Motion (FM), Out-of-View (OV), Background Clutter (BC), and Motion Blur (MB). Fig. 8 illustrates the sequence of numbers associated with each of the 11 challenges.

The experiments were conducted utilizing PyTorch on a system featuring an RTX 4070 GPU. The model underwent training for 20 iterations in total utilizing the stochastic gradient descent optimizer, with the learning rate fixed at 0.001. The size of the training batch was configured to be 76. For the first 10 iterations, the parameters are frozen to train the attention and head networks, followed by 10 main training iterations for the modified GoogLeNet [53,54]. In addition, we selected seven representative trackers SiamATT [40], SiamRAAN [41], TransT [32], ODTrack [33], BS-SiamRPN [16], ADSiamRPN [18], and SiamPKHT [19] for comparison.

4.2. Qualitative comparison

Figs. 10 to 14 show the performance of these advanced trackers in different scenarios. Compared to other trackers, our algorithm provides more reliable target recognition capabilities.

Fig. 10 illustrates the swift motion of the airplane, which encounters different levels of obstruction throughout its trajectory. when the object is partially occluded in frame #13, our tracker that integrates

the accumulated template is still able to accurately locate the target. Additionally, although most trackers can locate the target afterward, only our tracker maintains good overlap with the ground truth (frame #80).

In Fig. 11, the cyclist starts riding from a distance, gradually approaching the camera and becoming larger. In frames #63 to #84, the cyclist passes under a tree shadow, blending into the environment. SiamATT can only see part of the target, whereas our algorithm achieves perfect target localization because of retaining reliable spectral information in the band selection process. In frame #96, the target undergoes a significant deformation, and both our tracker and SiamPKHT correctly detect the target. It is worth mentioning that SiamTU notably excels in localizing the whole object more effectively.

As shown in Fig. 12, a ship is sailing at sea and encounters occlusion and interference from a similar ship during its journey. The SiamPKHT algorithm is misled and locks onto a similar ship, while SiamRAAN entirely loses track of the target. In frame #198, most trackers are positioned near the target. Nevertheless, the proposed algorithm effectively determines the exact location of the target.

In Fig. 13, a hand is rapidly moving a target bottle among several distracting objects in a complex background. During movement, the target is partially occluded at times, which presents a significant challenge for tracking. Most trackers fail to respond adequately to the changes in the target, with many losing the target's position. To our satisfaction, SiamTU successfully tracked the target throughout.

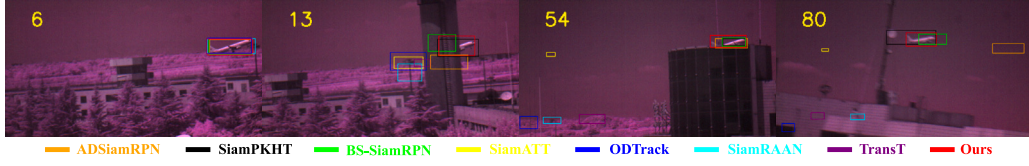


Fig. 10. Qualitative results on video airplane4.

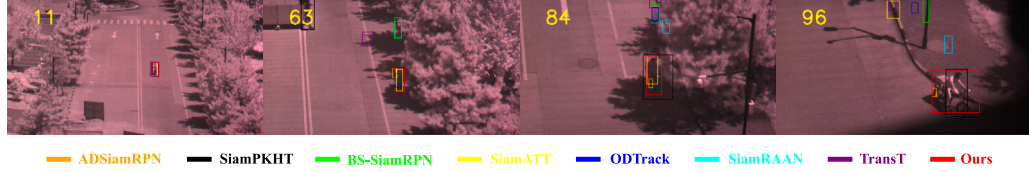


Fig. 11. Qualitative results on video bicycle.

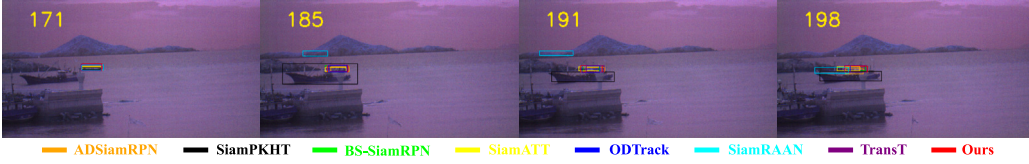


Fig. 12. Qualitative results on video boat6.

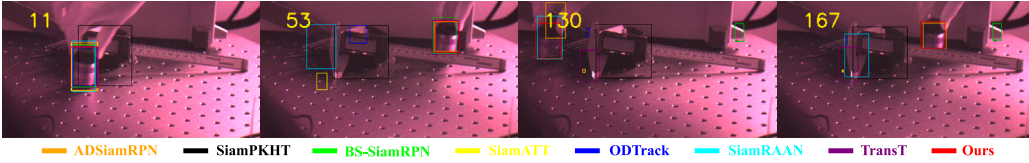


Fig. 13. Qualitative results on video camera.

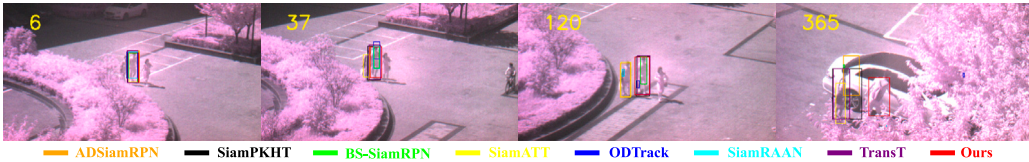


Fig. 14. Qualitative results on video triple.

Fig. 14 shows three individuals walking side by side on the road, moving from a distance to a closer position, and finally opening the doors of different cars. The action of opening the door causes the shape of the target to change. In addition, the target is partially occluded by tree branches to varying degrees. Most of the trackers correctly positioned the target in the early stage. However, from frames #37 to #120, ADSiamRPN, ODTrack, SiamRAAN, and SiamATT lose the target position, while BS-SiamRPN manages to partially locate the target. SiamPKHT, TransT, and SiamTU successfully find the target. In frame #365, despite interference from occlusion and deformation, only ours successfully obtains the accurate tracking result.

4.3. Quantitative comparison

Evaluation metrics comprise precision plot, success plot, distance precision, Area Under the Curve (AUC), and Frames per Second (FpS). The outcomes of the experiment are presented in Table 1 and Fig. 15.

Fig. 15 gives a comparison of the trackers throughout the dataset. It is evident that SiamTU achieves a high AUC score (0.617). Faced with challenges such as rapid target movement, occlusion, and a cluttered background, SiamRAAN and TransT struggle, while ODTrack and SiamATT fail to effectively handle interference from similar objects. Traditional Siamese trackers do not perform well in videos with spectral

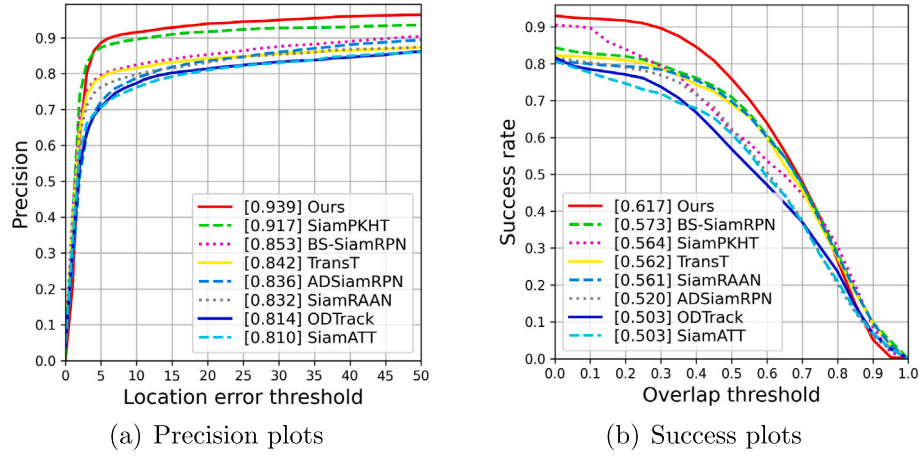


Fig. 15. The tracking evaluation that refer to 11 challenges. The specific metrics are detailed in Table 1.

Table 1

Performance comparisons (The red and green fonts indicate the best and second-best results, respectively. Location error threshold is set as 10).

Algorithm	Source	AUC ↑	Precision ↑	FpS ↑
MHT [11]	IEEE TIP	0.590	0.803	2.6
SiamATT [40]	Knowledge-based systems	0.503	0.810	50
TransT [32]	CVPR	0.562	0.842	75
BS-SiamRPN [16]	WHISPERS	0.573	0.853	70
ADSiamRPN [18]	Remote Sens.	0.520	0.846	45
SiamPKHT [19]	Sensors	0.564	0.917	68
SSTFT [34]	Remote Sens.	0.619	0.888	21
SiamBAG [28]	IEEE TGR	0.536	0.767	11
SPTrack [35]	Remote Sens.	0.588	0.927	24
ODTrack [33]	AAAI	0.503	0.814	51
SiamRAAN [41]	Neural Process. Lett.	0.561	0.832	60
SiamTU	Ours	0.617	0.939	65

information. SiamPKHT integrates pyramid shuffle attention to extract multiscale features and uses knowledge distillation to address the overfitting issue, thereby achieving good performance. Both SPtrack and SSTFT achieve good results because of the transformer. SiamBAG does not filter the spectral information, which may have caused interference from incorrect data. In contrast, the proposed algorithm introduces a band ranking strategy that can more effectively segment representative bands. In contrast to BS-SiamRPN, our algorithm integrates a template update module, which is advantageous in tackling challenges associated with OCC and SV.

Fig. 16 shows success rate plots referring to all attributes, while Fig. 17 presents precision plots. It can be seen that SiamTU exhibits good performance (AUC value is larger than 0.6) in the situation referring to appearance changes (DEF, FM, IPR, OPR, IV, MB, OV, SV), significantly outperforming other tracking algorithms in FM, IPR and SV. This results from the template update strategy, enabling the algorithm to more effectively adjust to the present condition of the target. In addition, SiamTU. SiamPKHT demonstrates a slightly higher success rate than SiamTU in both DEF and MB. This advantage is attributed to its pyramid shuffle attention module, which merges multiscale information to effectively capture variations in the target's appearance. However, in the remaining attributes, our algorithm surpasses SiamPKHT because we use band selection to process the input hyperspectral data, enabling the algorithm to focus on more valuable bands. BS-SiamRPN also achieved good results in LR. With regard to precision, SiamPKHT performs well in LR and OPR, while ADSiamRPN excels in IV. In general, our algorithm achieves high-level tracking across all attributes, demonstrating the strongest comprehensive performance.

Table 2

Ablation study of important components in SiamTU.

Experiment	BS	EMA	TU	AUC ↑	Precision ↑
I				0.515	0.719
II	✓			0.583	0.916
III		✓		0.507	0.697
IV			✓	0.511	0.784
V	✓	✓		0.589	0.924
VI		✓	✓	0.524	0.729
VII	✓		✓	0.596	0.934
VIII	✓	✓	✓	0.617	0.939

4.4. The ablation study

In the ablation study, we assess the performance of each component in SiamTU by employing AUC and precision as evaluation metrics. The reference method employs SiamGAT, as demonstrated in experiment I within Table 2. Furthermore, experiments are conducted on all frames to illustrate the incremental benefits of each component (BS, EMA, and TU) to the model's performance.

In experiment II (AUC: 0.583), the incorporation of BS into the baseline (0.515) significantly improves the results because BS effectively reduces spectral redundancy and retains relevant spectral information, allowing the algorithm to focus on meaningful information. Experiments III (0.507), IV (0.511), and VI (0.524) demonstrate that performance does not improve much without BS. From experiments V (0.589) and VII (0.596), it can be observed that applying EMA and TU to BS further enhances the tracker's performance. Finally, the experiment VIII (0.617) confirms the true effectiveness of the proposed method.

5. Conclusion

This study presents an improved Siamese approach aimed at tracking a hyperspectral object. The lightweight NGNMF model selects valuable bands for the Siamese model, combined with a feature refinement module to aggregate features, thus improving the hit rate of target feature matching. Moreover, the template update mechanism adapts well to target appearance change issues, thereby enhancing the performance of the tracker. In addition, extensive validation and ablation experiments on publicly available HSVs demonstrate that the SiamTU model excels at tracking a hyperspectral object. However, the proposed approach has certain limitations. The template update mechanism may pose challenges for real-time applications and may struggle with rapid and drastic appearance changes, potentially leading to tracking drift. Future work we will address these limitations to further improve the robustness and practicality of hyperspectral object tracking.

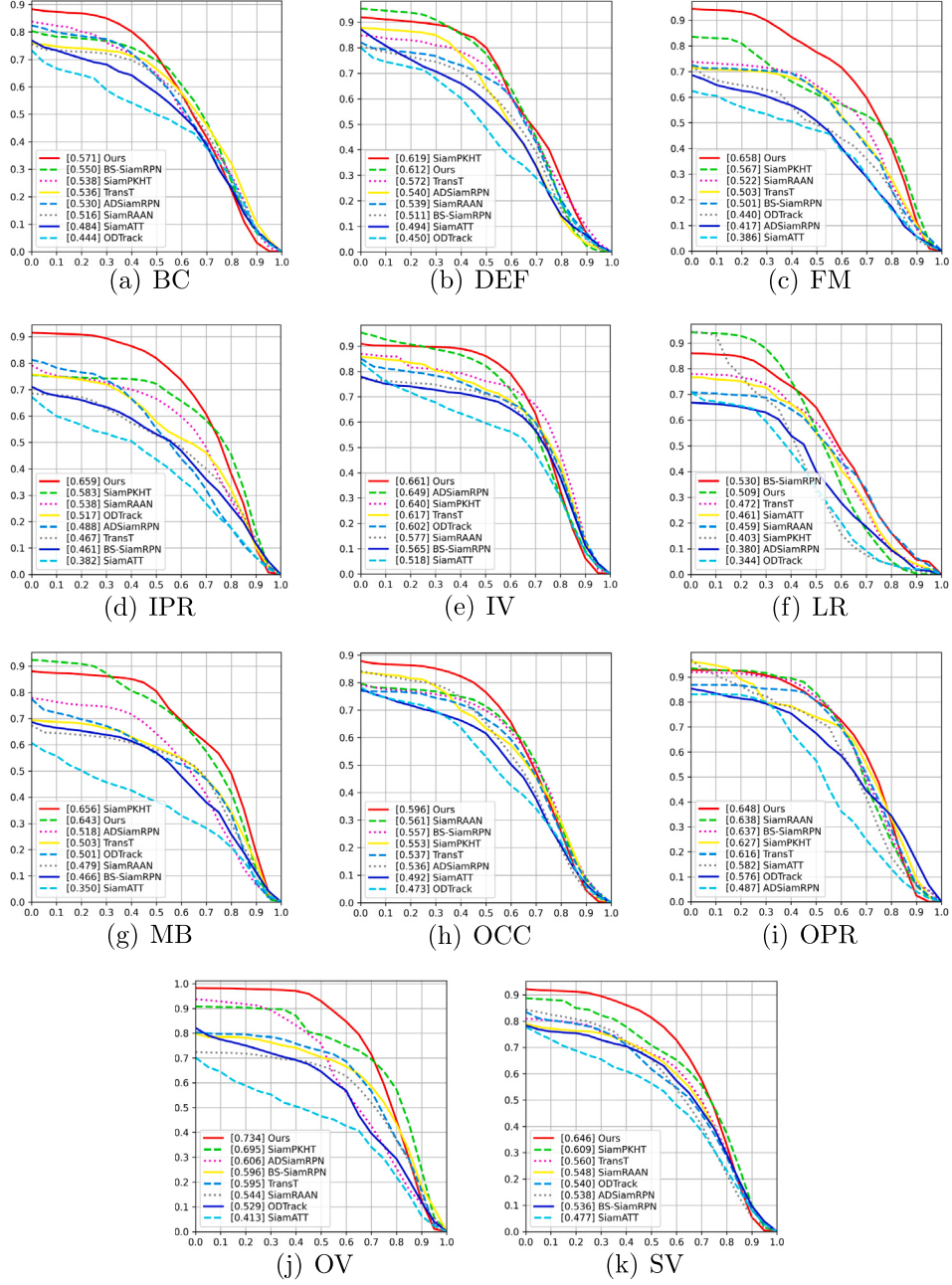


Fig. 16. Success plots of the IMEC25 dataset across 11 challenges.

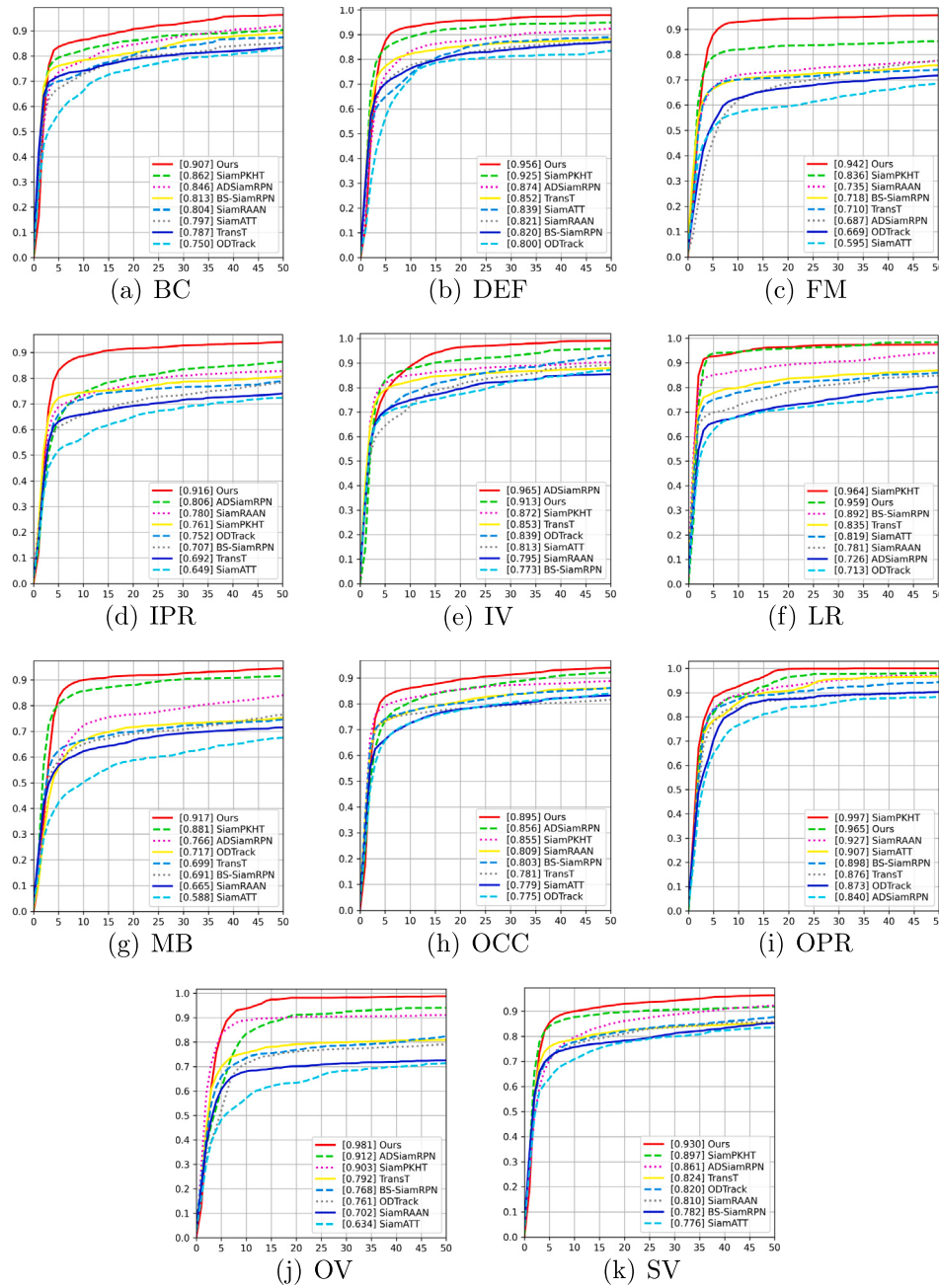


Fig. 17. Precision plots of the IMEC25 dataset across 11 challenges.

CRedit authorship contribution statement

Jialing Chen: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Data curation. **Kun Qian:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Wei Song:** Resources, Project administration, Formal analysis, Conceptualization. **Shiqing Wang:** Writing – review & editing, Supervision, Software, Methodology, Data curation. **Yonghao Qiu:** Methodology, Resources, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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